

CHE 201 Spring 2020 HW 2

Anna Christensen

TOTAL POINTS

74 / 100

QUESTION 1

1 Question 1 20 / 20

- ✓ - **0 pts** Correct
- **5 pts** Effect of friction is wrong or not answered.
- **5 pts** Calculation of constant force R is wrong.
- **5 pts** Calculation of the force of gravity is wrong.
- **5 pts** Energy balance is wrong.

QUESTION 2

2 Question 2 15 / 20

- **0 pts** Correct
- ✓ - **5 pts** Wrong calculation in b)
- ✓ - **0 pts** Be more precise on calculation. Keep more significant digits.
- **5 pts** Wrong equation in b)
- **5 pts** Wrong calculation in a)
- **5 pts** Wrong equation in a)

QUESTION 3

3 Question 3 4 / 20

- **0 pts** Correct
- ✓ - **4 pts** Wrong sketch of the processes.
- **4 pts** Wrong answer in a)
- ✓ - **4 pts** Wrong 1-2 work in b)
- ✓ - **4 pts** Wrong 2-3 work in b)
- **0 pts** Please distinguish a positive or a negative sign to denote the energy transfer direction by work.
- ✓ - **4 pts** Wrong 3-1 work in b)

QUESTION 4

4 Question 4 20 / 20

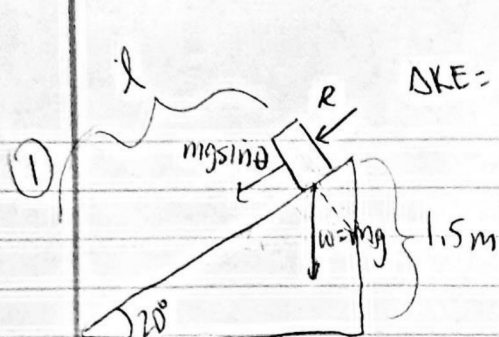
- ✓ - **0 pts** Correct
- **5 pts** Wrong equation used in getting power
- **5 pts** Wrong calculation in getting power.
- **5 pts** Wrong equation used in getting the speed

- **5 pts** Wrong calculation in getting the speed

QUESTION 5

5 Question 5 15 / 20

- **0 pts** Correct
- **2.5 pts** Not enough current federal regulations.
- **2.5 pts** Not enough current regulations in state and local area.
- ✓ - **5 pts** Not enough comments on the effectiveness of services.
- **20 pts** Question 5 not answered.
- **5 pts** No current federal regulations.
- **2 pts** No reference added.
- **5 pts** No current regulations in state or local area.



$$\Delta KE = \frac{1}{2} m (v_2^2 - v_1^2) = \int_{s_1}^{s_2} F ds$$

$$F = R + mgsin\theta$$

$$F = 0.05 N + (0.4 kg)(9.81 \frac{m}{s^2}) \sin(20)$$

$$F = 1.39 N$$

$$\text{length of incline} = \frac{1.5}{\sin 20}$$

$$l = \frac{1.5}{\sin 20}$$

$$l = 4.39 m$$

$$a = \frac{F}{m}$$

$$a = \frac{1.39 N}{0.4 kg} = 3.47 \frac{m}{s^2}$$

~~not~~

$$\Delta KE = \int_0^{4.39} 1.39 ds$$

$$\Delta KE = 1.39(4.39) = \boxed{6.1 J}$$

kinetic energy is increasing as the can moves down the slope.

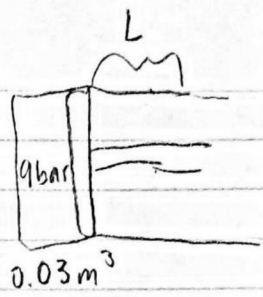
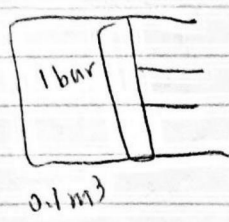
with friction, the force will have a lesser value because friction^{to u} is working in the opposite direction so we must subtract it. The smaller force means that the change in kinetic energy will be less.

1 Question 1 20 / 20

✓ - 0 pts Correct

- 5 pts Effect of friction is wrong or not answered.
- 5 pts Calculation of constant force R is wrong.
- 5 pts Calculation of the force of gravity is wrong.
- 5 pts Energy balance is wrong.

2

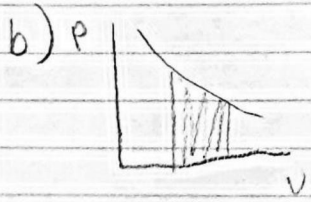


a) $p_1 = 1 \text{ bar}$ $p_2 = 9 \text{ bar}$ $pV^2 = \text{constant}$
 $V_1 = 0.1 \text{ m}^3$ $V_2 = ?$

$$1(0.1)^2 = 0.01$$

$$9(V_2)^2 = 0.01$$

$$V_2 = 0.03 \text{ m}^3$$



$$W = \int_{V_1}^{V_2} p \, dV$$

$$W = \int_{0.1}^{0.03} \frac{0.01}{V^2} \, dV$$

$$pV^2 = \text{constant}$$

$$pV^2 = 0.01$$

$$p = \frac{0.01}{V^2}$$

$$W = -2 \frac{0.01}{V^3} \Big|_{0.1}^{0.03} = -20 - (-741)$$

$$W = 721 \text{ J}$$

2 Question 2 15 / 20

- 0 pts Correct
- ✓ - 5 pts Wrong calculation in b)
- ✓ - 0 pts Be more precise on calculation. Keep more significant digits.
- 5 pts Wrong equation in b)
- 5 pts Wrong calculation in a)
- 5 pts Wrong equation in a)

③ process 1-2

$$p_1 = 10 \text{ lbf/in}^2$$

$$V_1 = 4 \text{ ft}^3$$

$$p_2 = 50 \text{ lbf/in}^2$$

$$V_2 = ?$$

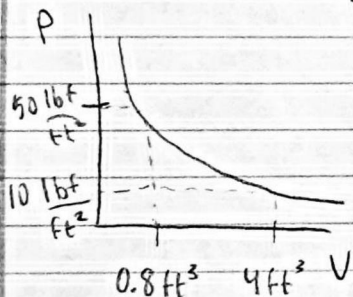
$$pV = \text{constant}$$

$$p_1 = 10 \frac{\text{lbf}}{\text{in}^2} \times \frac{12^2 \text{ in}^2}{1^2 \text{ ft}^2} = 1440 \frac{\text{lbf}}{\text{ft}^2}$$

$$p_2 = 50 \frac{\text{lbf}}{\text{in}^2} \times \frac{12^2 \text{ in}^2}{1^2 \text{ ft}^2} = 7200 \frac{\text{lbf}}{\text{ft}^2}$$

$$(1440 \frac{\text{lbf}}{\text{ft}^2})(4 \text{ ft}^3) = 5760 \text{ lbf} \cdot \text{ft}$$

$$5760 = (7200 \frac{\text{lbf}}{\text{ft}^2})(V_2) \rightarrow V_2 = 0.8 \text{ ft}^3$$



$$b) W = \int_{0.8}^4 \frac{5760}{V} dV$$

$$pV = 5760$$

$$p = \frac{5760}{V}$$

$$W = 5760 \left(\ln|V| \right) \Big|_{0.8}^4$$

$$W = 5760 (1.61) = 9270 \text{ J}$$

J $\rightarrow$ Btu?

$$9270 \text{ J} \times \frac{1 \text{ Btu}}{1055} = 8.79 \text{ Btu}$$

3 Question 3 4 / 20

- 0 pts Correct

✓ - 4 pts Wrong sketch of the processes.

- 4 pts Wrong answer in a)

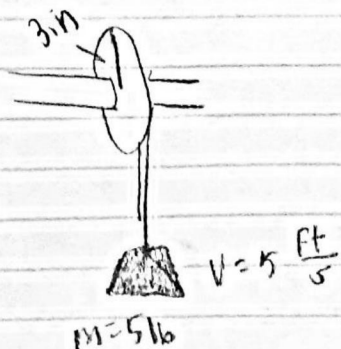
✓ - 4 pts Wrong 1-2 work in b)

✓ - 4 pts Wrong 2-3 work in b)

- 0 pts Please distinguish a positive or a negative sign to denote the energy transfer direction by work.

✓ - 4 pts Wrong 3-1 work in b)

④



$$a = g = 32.2 \frac{\text{ft}}{\text{s}^2}$$

$$V = 5 \frac{\text{ft}}{\text{s}}$$

a) power = $F \cdot V$

$$F = ma$$

$$m = 5 \text{ lb}$$

$$\text{power} = 5 \text{ lb} \cdot 5 \frac{\text{ft}}{\text{s}}$$

$$F = \frac{5 \text{ lb}}{32.2 \frac{\text{ft}}{\text{s}^2}} \cdot 32.2 \frac{\text{ft}}{\text{s}^2}$$

$$\frac{5 \text{ lb}}{32.2 \frac{\text{ft}}{\text{s}^2}}$$

$$\text{power} = 25 \frac{\text{lb} \cdot \text{ft}}{\text{s}}$$

$$F = 5 \text{ lbf}$$

$$25 \frac{\text{lb} \cdot \text{ft}}{\text{s}} \times \left(\frac{1 \text{ hp}}{550 \frac{\text{lb} \cdot \text{ft}}{\text{s}}} \right) = 0.0455 \text{ hp}$$

b) $\omega = \frac{V}{r}$

$$r = 3 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}} = 0.25 \text{ ft}$$

$$V = 5 \frac{\text{ft}}{\text{s}}$$

$$\omega = \frac{5 \frac{\text{ft}}{\text{s}}}{0.25 \text{ ft}} = 20 \frac{\text{rad}}{\text{sec}}$$

$$20 \frac{\text{rad}}{\text{sec}} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{60 \text{ sec}}{1 \text{ min}} = 19 \frac{\text{rev}}{\text{min}}$$

$$19 \frac{\text{rev}}{\text{min}}$$

4 Question 4 20 / 20

✓ - 0 pts Correct

- 5 pts Wrong equation used in getting power
- 5 pts Wrong calculation in getting power.
- 5 pts Wrong equation used in getting the speed
- 5 pts Wrong calculation in getting the speed

5)

United States federal law requires used nickel cadmium and lead batteries to be managed as “Universal Waster”. The Universal Waster rule prohibits handlers from throwing away nickel cadmium and lead batteries and requires them to be sent for recycling. (<https://www.call2recycle.org/recycling-laws-by-state/>). All lithium and lithium ion batteries must be bagged or taped at the ends to prevent fires.

Alkaline, non-rechargeable batteries do not require special disposal because they are not regulated as hazardous by Illinois state law. However, they are corrosive, and there are multiple locations available to recycle them. For example, one is called “Batteries Plus Bulbs” and another is “Interstate Batteries Center”. You can recycle these kinds of batteries there for a small fee. They can also be recycled through mail programs. (<https://ecologyactioncenter.org/hhw/battery-recycling/>) There are many more places throughout the state. Residents would need to look up a drop off location near them. I believe that this would discourage people from recycling their batteries by adding an extra step to the process. People would rather just throw them away rather than having to find a place to take the batteries and also find a way to transport them there. Also, the fact that people have to pay a fee to recycle will also discourage them.

Orland Fire Protection District advises residents to use electrical tape to cover old and unused batteries until they are properly recycled. They also say to store these batteries in dry, cool places. They listed multiple drop off locations in the area and listed batteries that are acceptable for recycling, such as: AAA, AA, C, D, hearing aid batteries, 6-volt, 9-volt, and home security batteries. Batteries that are not suitable for recycling include automobile, marine, and computer batteries, which must be dropped off at a local car service location. (<http://orlandfire.org/wp-content/uploads/home/RecycleYourBatteries.pdf>)

Orland Park waste management has a new service where they pick up electronics, motor oil, paint, batteries, and more at your door, so residents do not have to drive to a specific place to drop off used batteries. I think that this is very beneficial because it will encourage residents to properly dispose of their batteries since it will take less effort for them. (<https://orlandpark.org/services/waste-management>)

5 Question 5 15 / 20

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- 2.5 pts Not enough current regulations in state and local area.
- ✓ - 5 pts Not enough comments on the effectiveness of services.
- 20 pts Question 5 not answered.
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